

Square-Prefix Smooth–Transport Matching in Squared Complex Factor Space

Canonical cofactor parabolas, a moving cutoff,
and the $2ab$ lifetime–clustering mechanism

Fred Violen
OVVO Labs

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Abstract

Let

$$B_n = [n^2, (n+1)^2), \quad R_n = n+1, \quad X_n = R_n^2 - 1,$$

and define the complete square-prefix Mertens value

$$S_n = M(X_n) = \sum_{m \leq X_n} \mu(m).$$

At the exact square-root cutoff, every squarefree $m \leq X_n$ is uniquely either R_n -smooth or of the form cq with $c < R_n < q$ and q prime. This gives the unconditional identity

$$S_n = A_n - T_n.$$

The factor geometry is placed entirely in the squared complex plane. For a factor pair $cq = m$, define

$$a = \frac{c+q}{2}, \quad b = \frac{q-c}{2},$$

with signed b , and square

$$w = (a+bi)^2 = m + iY, \quad Y = 2ab = \frac{q^2 - c^2}{2}.$$

Writing $w = x + iy$ and $\rho = |w|$, the factors are recovered exactly by

$$c^2 = \rho - y, \quad q^2 = \rho + y.$$

Thus the squared point carries the source integer, the cofactor, the largest prime, the factor imbalance, the prefix-entry time, and the smoothness-transition time.

The paper distinguishes the *full factor fibre* of a fixed integer from the *canonical Möbius point* used by the smooth–transport decomposition. The full fibre contains every factor pair and has the universal odd-factor envelope

$$0 \leq Y \leq \frac{N^2 - 1}{2};$$

the familiar bound with smallest factor 3 is the nontrivial odd-factor envelope

$$Y \leq \frac{N^2 - 81}{18}.$$

The canonical point instead uses

$$q = P^+(m), \quad c = m/q,$$

so b and Y may be negative when the remaining cofactor exceeds the largest prime.

For fixed cofactor c , the canonical points lie on the parabola

$$\gamma_c(x) = \frac{x^2 - c^4}{2c^2},$$

subject to the exact arithmetic conditions that c is squarefree and $q = x/c$ is prime with $q > P^+(c)$. At cutoff R , the complete prefix is the vertical strip $x < R^2$, while the smooth-transport boundary is the moving parabola

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}.$$

The two parabolas meet exactly at

$$\gamma_c(x) = \Gamma_R(x) \iff x = cR,$$

and they meet orthogonally. Since an actual point on γ_c has $x = cq$, the intersection is precisely the transition $q = R$.

This yields an exact moving-region interpretation of $A_n - T_n$. A point enters the prefix when the vertical boundary reaches $x = m$, occupies the transport region while $\sqrt{m} < R < q$, and then crosses into the smooth region at $R = q$. The crossing mass appears with opposite signs in the increments of A and $-T$ and therefore telescopes identically. Only new points entering through the square-block strip contribute to the residual increment.

The same ordinate $Y = 2ab$ controls both spatial clustering and temporal persistence. For a canonical point in B_n , if $|Y| \leq H$, then the factor gap $|q - c|$ is at most H/n , and for each fixed gap there is at most one product in B_n . Hence

$$\#\{m \in B_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor,$$

with no points at all for $|Y_m| \leq n$. For transport points,

$$L(m) \leq q - \sqrt{cq} = \frac{2Y}{q(1 + \sqrt{c/q})(1 + c/q)} \leq \frac{2Y}{q}.$$

At the natural height $H \asymp n$, both block occupancy and transport lifetime are $O(1)$.

Finally, the quadratic map

$$\Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right)$$

is conformal: its differential expands both coordinate directions by $\sqrt{c^2 + q^2}$ and its Jacobian is

$$\det D\Phi = c^2 + q^2 \geq 2cq.$$

At square scale $cq \asymp n^2$, unit factor-lattice area expands by at least order n^2 , exactly the two-power dilution required to pass from the coherent N^5 ceiling to the cubic N^3 scale. Turning this geometric dilution into a signed high-shell kernel estimate is the remaining analytic step. The arithmetic decomposition, all coordinate identities, the moving-region partition, the channel telescoping law, the conformal Jacobian, and the low-height clustering and lifetime bounds are unconditional.

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1 Introduction and principal results

The square-prefix framework organizes the Mertens function at the complete boundaries

$$X_n = (n + 1)^2 - 1.$$

Its arithmetic advantage is that the cutoff has an exact square root. A squarefree integer below R^2 cannot contain two prime factors larger than R . Consequently, every nonsmooth source has a unique largest-prime representation $m = cq$ with

$$c < R < q.$$

This produces the exact decomposition

$$S_n = A_n - T_n.$$

The geometric question is why the two large components A_n and T_n match closely enough that their residual remains on the conjectural cubic energy scale. The answer developed here uses the squared complex factor coordinate

$$(a + bi)^2 = m + 2abi.$$

The key observation is not merely that the factor hyperbola becomes a vertical line. The entire smooth-transport process becomes a moving partition of a fixed signed point cloud:

1. each squarefree source has one canonical point in the squared complex plane;
2. fixed cofactors generate explicit parabolic curves;
3. the prefix cutoff is a moving vertical line;
4. the smoothness cutoff is a moving parabola;
5. every canonical point follows the state path

$$\text{outside} \longrightarrow \text{transport} \longrightarrow \text{smooth};$$

6. the transport-to-smooth crossing telescopes exactly;
7. the vertical coordinate $2ab$ measures both factor separation and transport lifetime;
8. the quadratic map expands factor-lattice area by $c^2 + q^2 \asymp n^2$ at square scale n .

The main exact results are summarized below.

Theorem 1.1 (Exact square-root decomposition). *For $R_n = n + 1$ and $X_n = R_n^2 - 1$, define*

$$A_n := \sum_{\substack{m \leq X_n \\ P^+(m) \leq R_n}} \mu(m),$$

$$T_n := \sum_{c < R_n} \mu(c) \left[\pi \left(\left\lfloor \frac{X_n}{c} \right\rfloor \right) - \pi(R_n) \right].$$

Then

$$\boxed{S_n = A_n - T_n.}$$

Theorem 1.2 (Squared-space recovery). *Let $w = x + iy$ be generated by a positive factor pair $cq = x$ through*

$$y = \frac{q^2 - c^2}{2}.$$

With $\rho = |w|$,

$$\boxed{c^2 = \rho - y, \quad q^2 = \rho + y.}$$

Hence the squared point determines the ordered factor pair exactly.

Theorem 1.3 (Cofactor-curve cutoff intersection). *For fixed $c > 0$, put*

$$\gamma_c(x) = \frac{x^2 - c^4}{2c^2}.$$

At cutoff $R > 0$, put

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}.$$

Then

$$\boxed{\gamma_c(x) = \Gamma_R(x) \iff x = cR.}$$

The curves meet orthogonally at their intersection. At an arithmetic point $x = cq$, the sign of $\gamma_c(cq) - \Gamma_R(cq)$ is the sign of $q - R$.

Theorem 1.4 (Exact channel telescoping). *As R increases, a canonical point enters the prefix at $R = \sqrt{cq}$ and crosses from transport to smooth at $R = q$. The signed mass crossing the moving parabola appears with opposite signs in the increments of A and $-T$. Therefore all internal channel transfers cancel from the increment of $A - T$; only points entering the new square strip remain.*

Theorem 1.5 (Low-height clustering and lifetime). *Let $m = cq \in B_n$ be canonical and let*

$$Y_m = \frac{q^2 - c^2}{2}.$$

Then, for $H \geq 0$,

$$\#\{m \in B_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor,$$

and the count is zero when $H = n$. For odd sources the bound improves to $\lfloor H/(2n) \rfloor$. If $Y_m > 0$, the transport lifetime satisfies

$$L(m) \leq \frac{2Y_m}{q}.$$

Consequently, for fixed Λ , the canonical points with $|Y_m| \leq \Lambda n$ have uniformly bounded block occupancy, and the transport members have uniformly bounded lifetime.

Theorem 1.6 (Major-arc rank-one law for cofactor blocks). *Fix $\eta > 0$ and a dyadic cofactor scale $C \asymp N^\alpha$ with $0 \leq \alpha \leq 1 - \eta$. On the local square-prefix interval $N \leq n < 2N$, let*

$$B_C(n) := - \sum_{C < c \leq 2C} \mu(c) \Pi_c(X_n), \quad a_C := \sum_{C < c \leq 2C} \frac{\mu(c)}{c},$$

and let $P_n = -\pi(X_n)$ be the prime-curve vector. Then

$$\boxed{B_C(n) = \frac{2}{2 - \alpha} a_C P_n + O_\eta \left(\frac{N^2}{\log^2 N} \right)}$$

uniformly for $N \leq n < 2N$. Consequently, the local Gram matrix of any fixed collection of such cofactor blocks is a rank-one prime-curve main term plus an error of order $O_\eta(N^5 / \log^3 N)$ entrywise.

Theorem 1.7 (Quadratic conformal expansion). *The map*

$$\Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right)$$

has

$$D\Phi(c, q)^\top D\Phi(c, q) = (c^2 + q^2)I,$$

and

$$\det D\Phi(c, q) = c^2 + q^2 \geq 2cq.$$

At square scale $cq \asymp n^2$, lengths expand by at least order n and areas by at least order n^2 .

The remaining theorem is an operator estimate: prove that the high-height, cross-curve signed interactions inherit this geometric dilution when inserted into the exact square-prefix kernel. No claim is made that a coordinate change alone proves cancellation; the point is that the coordinate change exposes the exact two-power scale and localizes the unresolved interaction.

2 Square prefixes and the exact arithmetic split

Define the complete square block

$$B_n := \{m \in \mathbb{N} : n^2 \leq m < (n+1)^2\}, \quad (1)$$

the square-root cutoff

$$R_n := n + 1, \quad (2)$$

and the complete-prefix endpoint

$$X_n := R_n^2 - 1 = (n+1)^2 - 1. \quad (3)$$

The square-prefix Mertens value is

$$S_n := M(X_n) = \sum_{m \leq X_n} \mu(m). \quad (4)$$

Because $\mu(R_n^2) = 0$ for $R_n > 1$, one may equally write $S_n = M(R_n^2)$.

Write $P^+(m)$ for the largest prime factor of m , with $P^+(1) = 1$.

Definition 2.1 (Smooth and transport terms). Define

$$A_n := \sum_{\substack{1 \leq m \leq X_n \\ P^+(m) \leq R_n}} \mu(m), \quad (5)$$

$$T_n := \sum_{1 \leq c < R_n} \mu(c) \left[\pi\left(\left\lfloor \frac{X_n}{c} \right\rfloor\right) - \pi(R_n) \right]. \quad (6)$$

Lemma 2.2 (One-large-prime closure). *Let $m \leq R^2 - 1$ be squarefree. If $P^+(m) > R$, then there is a unique factorization*

$$m = cq, \quad 1 \leq c < R < q, \quad q \text{ prime}, \quad (7)$$

where $q = P^+(m)$. Moreover,

$$\mu(m) = -\mu(c). \quad (8)$$

Proof. Put $q = P^+(m)$ and $c = m/q$. If $c \geq R$, then $m = cq > R^2$, contrary to $m \leq R^2 - 1$. Since m is squarefree, $q \nmid c$ and $\mu(cq) = -\mu(c)$. Uniqueness follows from the largest-prime choice. $\square$

Proof of Theorem 1.1. Partition the squarefree $m \leq X_n$ according to whether $P^+(m) \leq R_n$ or $P^+(m) > R_n$. The first population contributes A_n . By Theorem 2.2, the second population is uniquely $m = cq$ with $c < R_n < q$, and each source has weight $-\mu(c)$. The total second contribution is therefore $-T_n$. $\square$

2.1 The exact largest-prime cofactor identity

For $c \geq 1$, define the prime-tail counting function

$$\Pi_c(x) := \# \left\{ q \text{ prime} : P^+(c) < q \leq \left\lfloor \frac{x}{c} \right\rfloor \right\}. \quad (9)$$

Equivalently,

$$\Pi_c(x) = \left(\pi\left(\left\lfloor \frac{x}{c} \right\rfloor\right) - \pi(P^+(c)) \right)_+, \quad (10)$$

where $(u)_+ := \max\{u, 0\}$. The positive part is essential when $x/c < P^+(c)$.

Theorem 2.3 (Exact largest-prime cofactor expansion). *For every real $x \geq 1$,*

$$M(x) = 1 - \sum_{c \geq 1} \mu(c) \Pi_c(x). \quad (11)$$

The sum is finite, and separating the prime curve $c = 1$ gives

$$M(x) = 1 - \pi(x) - \sum_{2 \leq c \leq x/2} \mu(c) \left(\pi\left(\left\lfloor \frac{x}{c} \right\rfloor\right) - \pi(P^+(c)) \right)_+. \quad (12)$$

Proof. Every squarefree $m > 1$ has a unique largest prime factor $q = P^+(m)$. Writing $c = m/q$, one has

$$q > P^+(c), \quad \mu(m) = \mu(cq) = -\mu(c).$$

Conversely, every pair consisting of a squarefree c and a prime $q > P^+(c)$ produces the squarefree integer $m = cq$ whose largest prime factor is q . The condition $m \leq x$ is exactly $q \leq x/c$. Summing $-\mu(c)$ over these unique pairs proves Equation (11). The $c = 1$ term is $-\pi(x)$, because $P^+(1) = 1$. No term with $c > x/2$ can occur. $\square$

Remark 2.4 (Central analytic engine). Equation (12) is the exact arithmetic form of the cofactor-parabola decomposition. It shows that the Mertens function is not obtained by proving that the prime curve is small. The prime curve $-\pi(x)$ is cancelled by the Möbius-weighted collection of all curves $c \geq 2$. The geometric and large-sieve parts of the paper are methods for controlling the rate of this exact cancellation on square prefixes.

2.2 An unconditional major-arc rank-one theorem

The first analytic component of the cofactor cascade can be proved unconditionally. It explains why the small-cofactor Gram matrix is nearly rank one and identifies the scalar coefficients governing its large coherent mode.

For a dyadic block $C < c \leq 2C$, define

$$B_C(n) := - \sum_{C < c \leq 2C} \mu(c) \Pi_c(X_n), \quad a_C := \sum_{C < c \leq 2C} \frac{\mu(c)}{c}, \quad (13)$$

and write

$$P_n := -\pi(X_n). \quad (14)$$

Theorem 2.5 (Polynomial-range major-arc rank-one law). *Fix $\eta > 0$. Let N tend to infinity, let $0 \leq \alpha \leq 1 - \eta$, and choose a dyadic C satisfying $C \asymp N^\alpha$. Uniformly for $N \leq n < 2N$,*

$$\boxed{B_C(n) = \lambda_\alpha a_C P_n + O_\eta\left(\frac{N^2}{\log^2 N}\right), \quad \lambda_\alpha := \frac{2}{2 - \alpha}.} \quad (15)$$

Consequently,

$$\sum_{N \leq n < 2N} |B_C(n) - \lambda_\alpha a_C P_n|^2 \ll_\eta \frac{N^5}{\log^4 N}. \quad (16)$$

If $D \asymp N^\beta$ with $0 \leq \beta \leq 1 - \eta$, then

$$\boxed{\langle B_C, B_D \rangle_{[N, 2N]} = \lambda_\alpha \lambda_\beta a_C a_D \|P\|_{[N, 2N]}^2 + O_\eta\left(\frac{N^5}{\log^3 N}\right).} \quad (17)$$

Proof. For $c \asymp N^\alpha$ and $N \leq n < 2N$, one has $X_n/c \gg_\eta N^{1+\eta}$, while $P^+(c) \leq c \ll N^{1-\eta}$. Thus, for sufficiently large N , the positive part in $\Pi_c(X_n)$ is inactive and

$$\Pi_c(X_n) = \pi(X_n/c) - \pi(P^+(c)).$$

The prime number theorem with its standard first remainder term gives, uniformly in this range,

$$\pi(X_n/c) = \frac{X_n/c}{\log(X_n/c)} + O_\eta\left(\frac{X_n}{c \log^2 N}\right),$$

and

$$\pi(X_n) = \frac{X_n}{\log X_n} + O\left(\frac{X_n}{\log^2 N}\right).$$

Since

$$\frac{\log X_n}{\log(X_n/c)} = \frac{2}{2 - \alpha} + O_\eta\left(\frac{1}{\log N}\right),$$

it follows that

$$\pi(X_n/c) = \frac{\lambda_\alpha}{c} \pi(X_n) + O_\eta\left(\frac{N^2}{c \log^2 N}\right).$$

Summing the error uses $\sum_{C < c \leq 2C} c^{-1} \ll 1$. The fixed lower-prime exclusions satisfy

$$\sum_{C < c \leq 2C} \pi(P^+(c)) \leq \sum_{C < c \leq 2C} c \ll C^2 \ll_\eta N^{2-2\eta},$$

which is absorbed by $N^2/\log^2 N$. This proves Equation (15). Squaring and summing over N local prefixes gives Equation (16). Finally, write each block as its prime-vector main term plus its error and apply Cauchy–Schwarz, using $\|P\|_{[N,2N)} \asymp N^{5/2}/\log N$. $\square$

Corollary 2.6 (Rank-one major-arc Gram matrix). *For finitely many polynomial cofactor blocks with exponents bounded by $1 - \eta$, their local Gram matrix has the form*

$$G_N = \|P\|_{[N,2N)}^2 vv^\top + E_N, \quad v_j = \lambda_{\alpha_j} a_{C_j}, \quad (18)$$

with every entry of E_N bounded by $O_\eta(N^5/\log^3 N)$. Thus the coherent major-arc component cancels between dyadic cofactor blocks entirely through the scalar harmonic Möbius coefficients a_{C_j} , before any minor-arc estimate is used.

Remark 2.7 (What this theorem accomplishes). Theorem 2.5 is a genuine major-arc cancellation theorem for substantial ranges $C \asymp N^\alpha$, $\alpha < 1$. It is not RH: it does not control cofactor scales $C \gtrsim N$, nor does it supply a power saving for the rank-one remainder. It does prove that the large small-cofactor interactions are analytically organized rather than arbitrary, and it identifies the exact main term that the Squared-Cofactor Large Sieve must remove.

For

$$\mathbf{A}_N = (A_1, \dots, A_N)^\top, \quad \mathbf{T}_N = (T_1, \dots, T_N)^\top, \quad \mathbf{S}_N = (S_1, \dots, S_N)^\top,$$

we have

$$\mathbf{S}_N = \mathbf{A}_N - \mathbf{T}_N \quad (19)$$

and

$$\sum_{n \leq N} |S_n|^2 = \|\mathbf{A}_N - \mathbf{T}_N\|_2^2. \quad (20)$$

3 The full factor fibre of a fixed integer

The factorization geometry and the smooth–transport geometry use the same coordinate map but different point selections. This distinction is essential.

Fix $N \geq 1$ and consider every positive factor pair

$$uv = N, \quad 1 \leq u \leq v. \quad (21)$$

Define the Fermat coordinates

$$a = \frac{u+v}{2}, \quad b = \frac{v-u}{2} \geq 0. \quad (22)$$

Then

$$u = a - b, \quad v = a + b, \quad N = a^2 - b^2. \quad (23)$$

The coordinates lie in $\frac{1}{2}\mathbb{Z}$; they are integers when u and v have the same parity.

Squaring the complex factor gives

$$(a + bi)^2 = N + iY, \quad \boxed{Y = 2ab = \frac{v^2 - u^2}{2}}. \quad (24)$$

Thus all factorizations of a fixed N lie on the vertical fibre

$$\operatorname{Re} w = N. \quad (25)$$

3.1 Exact square-root tests

Let

$$w = N + iY, \quad \rho = |w| = \sqrt{N^2 + Y^2}. \quad (26)$$

Then

$$a^2 = \frac{\rho + N}{2}, \quad (27)$$

$$b^2 = \frac{\rho - N}{2}, \quad (28)$$

$$u^2 = \rho - Y, \quad (29)$$

$$v^2 = \rho + Y. \quad (30)$$

Accordingly, a candidate vertical ordinate Y corresponds to an exact factorization if and only if the relevant square and parity conditions hold. The identity

$$\sqrt{N^2 + Y^2} = a^2 + b^2 \quad (31)$$

is the right-triangle form of the same condition.

3.2 The full admissible intervals

For a factor pair $u(N/u) = N$ with $1 \leq u \leq \sqrt{N}$,

$$a(u) = \frac{u + N/u}{2}, \quad b(u) = \frac{N/u - u}{2}. \quad (32)$$

Both functions decrease as u increases toward $\sqrt{N}$. Therefore the full positive factor fibre satisfies

$$\sqrt{N} \leq a \leq \frac{N+1}{2}, \quad (33)$$

$$0 \leq b \leq \frac{N-1}{2}, \quad (34)$$

$$0 \leq Y \leq \frac{N^2-1}{2}. \quad (35)$$

For integer enumeration one uses $a \geq \lceil \sqrt{N} \rceil$ with the parity required by N .

The upper endpoint corresponds to the trivial factor pair $1 \cdot N$. If the trivial factor is excluded and the smallest permitted odd factor is 3, then

$$a \leq \frac{N+9}{6}, \quad (36)$$

$$b \leq \frac{N-9}{6}, \quad (37)$$

$$Y \leq \frac{N^2-81}{18}. \quad (38)$$

These are the nontrivial odd-factor or semiprime-search envelopes. They are not the universal bounds for the complete factor fibre.

Example 3.1 (The fibre $N = 309$). The trivial factor pair $1 \cdot 309$ gives

$$(a, b) = (155, 154), \quad w = 309 + 47740i.$$

The nontrivial factorization $3 \cdot 103$ gives

$$(a, b) = (53, 50), \quad w = 309 + 5300i.$$

The latter is the highest point allowed by the nontrivial odd-factor condition $u \geq 3$, but not the highest point on the full factor fibre.

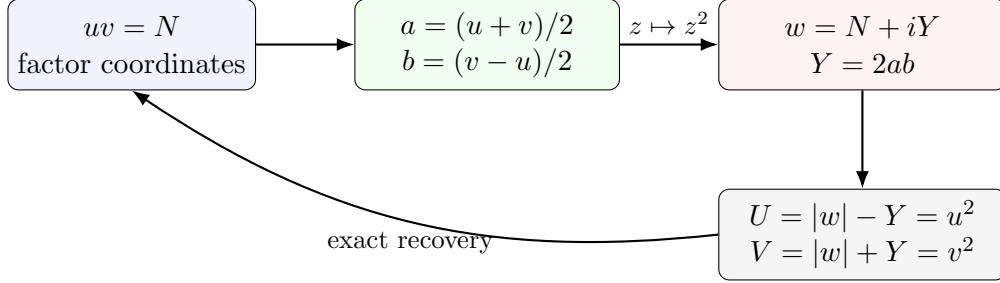


Figure 1: Equivalent coordinate systems on a complete factor fibre. The squared coordinates retain the factor pair exactly.

4 The canonical Möbius point cloud

The smooth-transport decomposition does not use every point on each factor fibre. It selects one canonical point per squarefree source.

For squarefree $m > 1$, define

$$q_m := P^+(m), \quad c_m := \frac{m}{q_m}. \quad (39)$$

Unlike the ordered factor pair in [Section 3](#), the canonical pair is not reordered. The cofactor c_m may exceed q_m . For example,

$$m = 30, \quad q_m = 5, \quad c_m = 6.$$

Therefore define the signed Fermat coordinate

$$a_m := \frac{c_m + q_m}{2}, \quad b_m := \frac{q_m - c_m}{2} \in \frac{1}{2}\mathbb{Z}, \quad (40)$$

and the canonical squared point

$$w_m := (a_m + ib_m)^2 = m + iY_m, \quad (41)$$

where

$$Y_m = 2a_mb_m = \frac{q_m^2 - c_m^2}{2}. \quad (42)$$

The sign of Y_m records the order of the canonical factors:

$$Y_m > 0 \iff c_m < q_m, \quad Y_m < 0 \iff c_m > q_m. \quad (43)$$

4.1 Exact parameterization by cofactors and primes

Every squarefree $m > 1$ has a unique representation

$$m = cq, \quad (44)$$

where

$$\mu(c) \neq 0, \quad q \text{ is prime}, \quad q > P^+(c). \quad (45)$$

Conversely, every pair satisfying Equation (45) produces a squarefree integer $m = cq$ with largest prime factor q . Moreover,

$$\mu(cq) = -\mu(c). \quad (46)$$

Define the canonical index set

$$\mathcal{C} := \{(c, q) : \mu(c) \neq 0, \ q \text{ prime}, \ q > P^+(c)\}. \quad (47)$$

The squared point cloud is

$$\mathcal{W} := \left\{ cq + i \frac{q^2 - c^2}{2} : (c, q) \in \mathcal{C} \right\}. \quad (48)$$

The signed point measure is

$$\nu := \delta_1 - \sum_{(c, q) \in \mathcal{C}} \mu(c) \delta_{cq + i(q^2 - c^2)/2}. \quad (49)$$

The atom at 1 records $\mu(1) = 1$.

Example 4.1 (A point below the real axis). For $m = 30$, the canonical pair is $(c, q) = (6, 5)$, so

$$a = \frac{11}{2}, \quad b = -\frac{1}{2}, \quad w_{30} = 30 - \frac{11}{2}i.$$

This point enters a square prefix only after the cutoff already exceeds $q = 5$, so it is smooth immediately and never occupies the transport channel.

5 Four exact coordinate systems

Let

$$w = x + iy = cq + i \frac{q^2 - c^2}{2}, \quad \rho := |w| = \sqrt{x^2 + y^2}. \quad (50)$$

Then

$$\rho = \frac{c^2 + q^2}{2}. \quad (51)$$

The most useful coordinate charts are collected in Table 1.

The central identities are

$$\boxed{U := \rho - y = c^2, \quad V := \rho + y = q^2.} \quad (52)$$

Consequently,

$$x = \sqrt{UV} = cq, \quad y = \frac{V - U}{2}. \quad (53)$$

Table 1: Equivalent coordinates for a canonical squared factor point.

Coordinate system	Definition	Exact inverse information
Factor coordinates	(c, q)	$m = cq, q = P^+(m)$
Fermat coordinates	$a = (c + q)/2, b = (q - c)/2$	$c = a - b, q = a + b$
Squared Cartesian	$x = cq, y = (q^2 - c^2)/2$	$w = x + iy = (a + ib)^2$
Radial-null	$U = \rho - y, V = \rho + y$	$U = c^2, V = q^2, x^2 = UV$
Square-root	$a^2 = (\rho + x)/2, b^2 = (\rho - x)/2$	sign of b is sign of y

These null coordinates linearize the arithmetic cutoffs. They also make the meaning of $2ab$ exact:

$$Y = 2ab = \frac{V - U}{2}, \quad (54)$$

so the imaginary coordinate is half the separation between the squared canonical factors.

6 Cofactor parabolas

Fix a positive cofactor c and let q vary. In squared Cartesian coordinates,

$$x = cq, \quad y = \frac{q^2 - c^2}{2}. \quad (55)$$

Eliminating $q = x/c$ gives the cofactor parabola

$$\gamma_c(x) = \frac{x^2 - c^4}{2c^2}. \quad (56)$$

The canonical point cloud is not the filled region between envelopes. It is the arithmetic sampling

$$x = cq, \quad q \text{ prime}, \quad q > P^+(c), \quad (57)$$

on the family [Equation \(56\)](#).

The curve crosses the real axis at

$$x = c^2. \quad (58)$$

Points to the right have $q > c$ and positive ordinate; points to the left have $q < c$ and negative ordinate.

The $c = 1$ curve is

$$\gamma_1(x) = \frac{x^2 - 1}{2}. \quad (59)$$

Its arithmetic samples are exactly the prime points, since $m = 1 \cdot q = q$. It is the outer positive envelope of the complete canonical cloud. The curve

$$\gamma_3(x) = \frac{x^2 - 81}{18} \quad (60)$$

is the first nontrivial odd-cofactor envelope used in semiprime search.

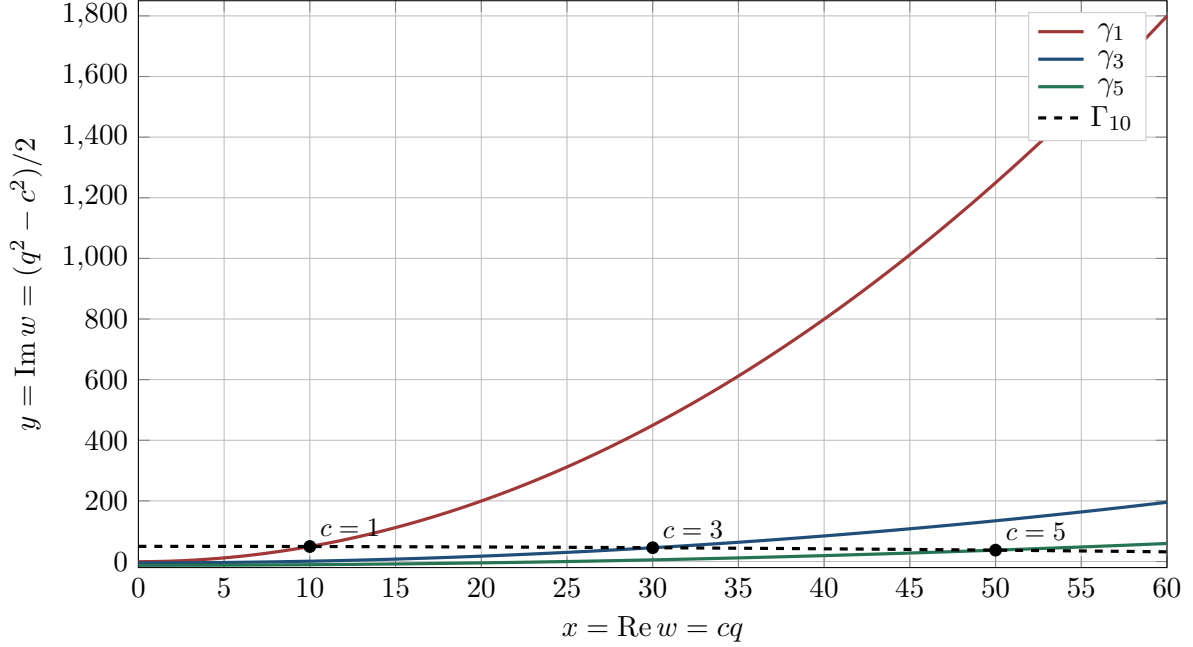


Figure 2: Cofactor parabolas γ_c and the moving cutoff parabola Γ_R for $R = 10$. Each γ_c meets Γ_R at $x = cR$. Actual canonical points are discrete prime samples on the curves.

7 The moving prefix and smoothness cutoff

Fix $R > 0$. The complete arithmetic prefix is

$$\mathcal{P}_R := \{w = x + iy \in \mathcal{W} : x < R^2\}. \quad (61)$$

Since $V = q^2 = |w| + y$, the smooth condition $q \leq R$ is

$$|w| + y \leq R^2. \quad (62)$$

Define

$$\mathcal{A}_R := \{w \in \mathcal{P}_R : |w| + \text{Im } w \leq R^2\}, \quad (63)$$

$$\mathcal{U}_R := \{w \in \mathcal{P}_R : |w| + \text{Im } w > R^2\}. \quad (64)$$

The prefix is the disjoint union

$$\mathcal{P}_R = \mathcal{A}_R \dot{\cup} \mathcal{U}_R. \quad (65)$$

Solving Equation (62) on the boundary gives

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}. \quad (66)$$

Inside the prefix strip $x < R^2$, the smooth region lies below the graph and the transport region lies above it.

In null coordinates, the same regions are even simpler:

$$\mathcal{P}_R : \quad UV < R^4, \quad (67)$$

$$\mathcal{A}_R: \quad UV < R^4, \quad V \leq R^2, \quad (68)$$

$$\mathcal{U}_R: \quad UV < R^4, \quad V > R^2. \quad (69)$$

For a transport point, $UV < R^4 < V^2$, so automatically $U < R^2$. Thus

$$\mathcal{U}_R: \quad U < R^2 < V, \quad UV < R^4. \quad (70)$$

This is exactly $c < R < q$.

7.1 Exact intersection with every cofactor curve

Proposition 7.1 (Intersection factorization). *For $c, R > 0$,*

$$\gamma_c(x) - \Gamma_R(x) = \frac{(R^2 + c^2)(x^2 - c^2 R^2)}{2c^2 R^2}. \quad (71)$$

Consequently,

$$\gamma_c(x) = \Gamma_R(x) \iff x = cR. \quad (72)$$

At the arithmetic point $x = cq$,

$$\gamma_c(cq) - \Gamma_R(cq) = \frac{(R^2 + c^2)(q^2 - R^2)}{2R^2}. \quad (73)$$

The sign is therefore exact:

$$q < R \iff w_{c,q} \text{ lies below } \Gamma_R, \quad (74)$$

$$q > R \iff w_{c,q} \text{ lies above } \Gamma_R. \quad (75)$$

Proposition 7.2 (Orthogonal crossing). *At the intersection $x = cR$,*

$$\gamma'_c(cR) = \frac{R}{c}, \quad \Gamma'_R(cR) = -\frac{c}{R}, \quad (76)$$

so

$$\gamma'_c(cR)\Gamma'_R(cR) = -1. \quad (77)$$

Hence each cofactor curve meets the moving cutoff orthogonally.

The transition is therefore transverse, never tangent. A canonical point cannot remain trapped in a thin boundary layer as the cutoff passes its largest prime.

8 The exact moving-region form of $A - T$

Using the signed measure [Equation \(49\)](#), define

$$\mathcal{A}(R) := \nu(\mathcal{A}_R), \quad \mathcal{U}(R) := \nu(\mathcal{U}_R). \quad (78)$$

At $R = R_n$,

$$\mathcal{A}(R_n) = A_n, \quad \mathcal{U}(R_n) = -T_n. \quad (79)$$

Indeed, the transport points satisfy $c < R_n < q$ and carry signed point mass

$$\mu(cq) = -\mu(c).$$

Therefore

$$\boxed{S_n = \nu(\mathcal{P}_{R_n}) = \mathcal{A}(R_n) + \mathcal{U}(R_n) = A_n - T_n.} \quad (80)$$

Equivalently, define projectors

$$P_R(w) := \mathbf{1}_{\{\operatorname{Re} w < R^2\}}, \quad (81)$$

$$Q_R(w) := \mathbf{1}_{\{|w| + \operatorname{Im} w \leq R^2\}}. \quad (82)$$

Then

$$\mathcal{A}(R) = \int P_R Q_R d\nu, \quad (83)$$

$$\mathcal{U}(R) = \int P_R (1 - Q_R) d\nu, \quad (84)$$

$$S(R) = \int P_R d\nu. \quad (85)$$

The identity is the pointwise partition

$$P_R = P_R Q_R + P_R (1 - Q_R). \quad (86)$$

9 Entry time, transition time, and exact telescoping

For a canonical source $m = cq$, define its first square-prefix index

$$e(m) := \lfloor \sqrt{m} \rfloor. \quad (87)$$

This is the least n such that $m < R_n^2$. Define the smoothness-transition index

$$h(m) := q - 1. \quad (88)$$

This is the least n such that $q \leq R_n$.

The transport lifetime is

$$L(m) := (h(m) - e(m))_+. \quad (89)$$

There are two cases:

1. If $q > \sqrt{m}$, equivalently $q > c$ and $Y > 0$, then the point enters above the moving parabola and occupies transport for the discrete interval

$$e(m) \leq n < h(m).$$

2. If $q \leq \sqrt{m}$, equivalently $q \leq c$ and $Y \leq 0$, then $h(m) \leq e(m)$ and the point enters already smooth. Its transport interval is empty.

Define the pointwise indicators

$$p_n(m) := \mathbf{1}_{\{e(m) \leq n\}}, \quad (90)$$

$$u_n(m) := \mathbf{1}_{\{e(m) \leq n < h(m)\}}, \quad (91)$$

$$a_n(m) := \mathbf{1}_{\{e(m) \leq n, h(m) \leq n\}}. \quad (92)$$

Then

$$\boxed{p_n(m) = u_n(m) + a_n(m)}. \quad (93)$$

9.1 Crossing cancellation

Let E_n^A be the signed mass entering the new vertical strip already below the moving parabola, let E_n^U be the signed mass entering above it, and let C_n be the signed mass crossed by the moving parabola between R_{n-1} and R_n . Then

$$\mathcal{A}(R_n) - \mathcal{A}(R_{n-1}) = E_n^A + C_n, \quad (94)$$

$$\mathcal{U}(R_n) - \mathcal{U}(R_{n-1}) = E_n^U - C_n. \quad (95)$$

Therefore

$$\boxed{S_n - S_{n-1} = E_n^A + E_n^U = \sum_{m \in B_n} \mu(m)}. \quad (96)$$

The entire crossing mass cancels.

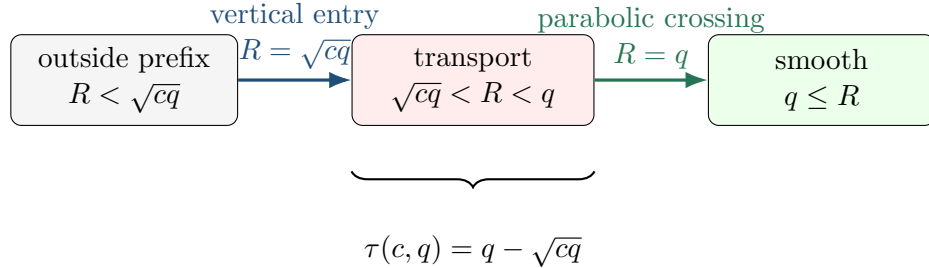


Figure 3: The exact state path of a positive-height canonical point. The second transition only reallocates signed mass between $-T$ and A , so it cancels from the residual increment.

This is the telescopic nature of the squared coordinate: the point's imaginary height determines when the moving parabola reaches it, while the crossing itself contributes zero to the increment of the total prefix mass.

10 The exact *2ab* lifetime law

Define the signed continuous delay

$$\tau(c, q) := q - \sqrt{cq}. \quad (97)$$

It is positive exactly for transport-capable points, zero at $c = q$, and negative for points that enter already smooth.

Put

$$r := \frac{c}{q}. \quad (98)$$

Then

$$Y = \frac{q^2}{2}(1 - r^2) \quad (99)$$

and

$$\tau = q(1 - \sqrt{r}). \quad (100)$$

Since

$$1 - r^2 = (1 - \sqrt{r})(1 + \sqrt{r})(1 + r), \quad (101)$$

we obtain an exact identity.

Theorem 10.1 (Exact height–lifetime identity). *For every positive pair (c, q) ,*

$$Y = \frac{q\tau}{2} \left(1 + \sqrt{\frac{c}{q}}\right) \left(1 + \frac{c}{q}\right). \quad (102)$$

Equivalently,

$$\tau = \frac{2Y}{q \left(1 + \sqrt{c/q}\right) (1 + c/q)}. \quad (103)$$

The sign of Y is therefore exactly the sign of the continuous delay. On the transport side $0 < c/q < 1$, the denominator in Equation (103) lies between q and $4q$, giving

$$\frac{Y}{2q} \leq \tau \leq \frac{2Y}{q}. \quad (104)$$

Since

$$L(m) = q - 1 - \lfloor \sqrt{m} \rfloor \leq q - \sqrt{m} = \tau(c, q) \quad (105)$$

for transport points,

$$L(m) \leq \frac{2Y_m}{q_m}. \quad (106)$$

In squared-space coordinates, $q = \sqrt{|w| + \operatorname{Im} w}$, so

$$L(w) \leq \frac{2(\operatorname{Im} w)_+}{\sqrt{|w| + \operatorname{Im} w}}. \quad (107)$$

A point close to the real axis cannot remain in transport for many prefix steps.

11 Low-height clustering in real square strips

The n th square block becomes the vertical strip

$$\mathcal{B}_n := \{w_m : n^2 \leq \operatorname{Re} w_m < (n+1)^2\}. \quad (108)$$

For a canonical pair (c, q) , define the absolute factor gap

$$d := |q - c|. \quad (109)$$

Then

$$|Y| = \frac{|q^2 - c^2|}{2} = \frac{d(q + c)}{2}. \quad (110)$$

For a squarefree source $m > 1$, one has $d \geq 1$: equality $d = 0$ would force $m = c^2$.

Lemma 11.1 (At most one product per positive absolute gap). *Fix $n \geq 1$ and an integer $d \geq 1$. There is at most one unordered positive factor pair $\{c, q\}$ with $|q - c| = d$ and*

$$n^2 \leq cq < (n + 1)^2. \quad (111)$$

Proof. Write the smaller factor as r and the larger as $r + d$. The product is $f_d(r) = r(r + d)$. If $f_d(r) \in B_n$, then

$$2r + d \geq 2\sqrt{r(r + d)} \geq 2n.$$

Hence

$$f_d(r + 1) - f_d(r) = 2r + d + 1 \geq 2n + 1,$$

which exceeds the integer diameter $2n$ of the block. Thus two distinct values of r cannot both occur. $\square$

Theorem 11.2 (Sharpened absolute low-height clustering). *For every $H \geq 0$,*

$$\boxed{\#\{w_m \in \mathcal{B}_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor}. \quad (112)$$

Moreover,

$$\boxed{\#\{w_m \in \mathcal{B}_n : |Y_m| \leq n\} = 0}. \quad (113)$$

Proof. If $cq \in B_n$, then

$$c + q \geq 2\sqrt{cq} \geq 2n.$$

Therefore

$$|Y| = \frac{d(c + q)}{2} \geq dn.$$

The condition $|Y| \leq H$ implies $d \leq \lfloor H/n \rfloor$. Since $d \geq 1$, there are at most $\lfloor H/n \rfloor$ possible gaps, and [Theorem 11.1](#) gives at most one point for each gap.

If $|Y| \leq n$, the inequalities force $d = 1$ and $c + q = 2n$. But $|q - c| = 1$ makes $c + q$ odd, whereas $2n$ is even. Hence equality is impossible and the set is empty. $\square$

11.1 Parity-refined bounds

The factor gap has a prescribed parity:

- if m is odd, then c and q are odd, so d is even;
- if m is even and squarefree, then the canonical factors have opposite parity, so d is odd.

Let

$$D := \left\lfloor \frac{H}{n} \right\rfloor. \quad (114)$$

The permitted positive gaps are therefore

$$2, 4, \dots, 2 \left\lfloor \frac{D}{2} \right\rfloor$$

for odd sources and

$$1, 3, \dots, 2 \left\lceil \frac{D}{2} \right\rceil - 1$$

for even sources. Consequently,

$$\#\{w_m \in \mathcal{B}_n : m \text{ odd}, |Y_m| \leq H\} \leq \left\lfloor \frac{H}{2n} \right\rfloor, \quad (115)$$

$$\#\{w_m \in \mathcal{B}_n : m \text{ even}, |Y_m| \leq H\} \leq \left\lceil \frac{1}{2} \left\lfloor \frac{H}{n} \right\rfloor \right\rceil. \quad (116)$$

At the endpoint $H = n$, both populations are empty by [Equation \(113\)](#).

At height $H = \Lambda n$, the scale-free bounds become

$$\#\{w_m \in \mathcal{B}_n : |Y_m| \leq \Lambda n\} \leq \lfloor \Lambda \rfloor, \quad (117)$$

$$\#\{m \text{ odd} : |Y_m| \leq \Lambda n\} \leq \left\lfloor \frac{\Lambda}{2} \right\rfloor. \quad (118)$$

11.2 Sharpened low-height space–time budget

For negative-height points, the transport lifetime is zero. For positive-height points in $\mathcal{B}_n$, one has $q \geq \sqrt{m} \geq n$, so [Equation \(106\)](#) gives

$$L(m) \leq \frac{2H}{n} \quad (0 < Y_m \leq H). \quad (119)$$

Therefore

$$\boxed{\sum_{\substack{w_m \in \mathcal{B}_n \\ |Y_m| \leq H}} (1 + L(m)) \leq \left\lfloor \frac{H}{n} \right\rfloor \left(1 + \frac{2H}{n}\right).} \quad (120)$$

The parity-refined versions are

$$\sum_{\substack{w_m \in \mathcal{B}_n \\ m \text{ odd}, |Y_m| \leq H}} (1 + L(m)) \leq \left\lfloor \frac{H}{2n} \right\rfloor \left(1 + \frac{2H}{n}\right), \quad (121)$$

$$\sum_{\substack{w_m \in \mathcal{B}_n \\ m \text{ even}, |Y_m| \leq H}} (1 + L(m)) \leq \left\lceil \frac{1}{2} \left\lfloor \frac{H}{n} \right\rfloor \right\rceil \left(1 + \frac{2H}{n}\right). \quad (122)$$

At $H = \Lambda n$, all three right sides are bounded independently of n .

12 The quadratic map is conformal and supplies two powers

Consider the continuous map

$$\Phi(c, q) := (x(c, q), y(c, q)) = \left(cq, \frac{q^2 - c^2}{2} \right). \quad (123)$$

Its differential is

$$D\Phi(c, q) = \begin{pmatrix} q & c \\ -c & q \end{pmatrix}. \quad (124)$$

The two columns are orthogonal and have equal length $\sqrt{c^2 + q^2}$. Consequently,

$$D\Phi(c, q)^\top D\Phi(c, q) = (c^2 + q^2)I. \quad (125)$$

Thus Φ is conformal: locally it rotates and expands every direction by the same factor

$$\lambda(c, q) = \sqrt{c^2 + q^2}. \quad (126)$$

The area Jacobian is

$$\boxed{J(c, q) = \det D\Phi(c, q) = c^2 + q^2 = 2|w|}. \quad (127)$$

By AM–GM,

$$c^2 + q^2 \geq 2cq = 2x. \quad (128)$$

Therefore, in the square strip $x \asymp n^2$,

$$\lambda(c, q) \gtrsim n, \quad J(c, q) \gtrsim n^2. \quad (129)$$

This is the two-dimensional version of the vertical spacing estimate. Unit-scale separation in the factor plane becomes scale- n separation in the squared plane, while unit factor-lattice area becomes scale- n^2 area. The coherent square-prefix ceiling loses exactly two powers if the relevant kernel neighborhoods have bounded squared-space area after localization.

Caution 12.1 (What the Jacobian does not prove). The Möbius point cloud is discrete and arithmetically restricted, and the square-prefix kernel is not Euclidean convolution. The Jacobian therefore does not by itself prove a signed variance estimate. It proves the exact geometric dilution scale that a valid operator argument must inherit.

12.1 Two-dimensional area versus one-dimensional cofactor curves

The conformal Jacobian governs two-dimensional regions in the continuous (c, q) factor plane. The arithmetic measure is not two-dimensional: for each fixed cofactor c , it is supported on the one-dimensional prime-sampled curve

$$q \mapsto \Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right), \quad q \text{ prime}, \quad q > P^+(c).$$

Along that curve,

$$\left\| \frac{\partial \Phi}{\partial q}(c, q) \right\| = \sqrt{c^2 + q^2}. \quad (130)$$

Thus one-dimensional arclength is also expanded, but the number and arithmetic placement of samples are governed by primes rather than by the area Jacobian. A two-dimensional density estimate cannot be applied directly to a single cofactor curve.

The outer curve $c = 1$ makes this obstruction explicit. Its contribution to the n th complete prefix is

$$S_n^{(1)} = -\pi(X_n), \quad (131)$$

because every prime $q \leq X_n$ has canonical pair $(1, q)$ and Möbius weight -1 . The prime number theorem therefore predicts

$$\sum_{n \leq N} |S_n^{(1)}|^2 \sim \frac{N^5}{20 \log^2 N}, \quad (132)$$

up to the usual slowly varying correction. The prime curve by itself is not on the cubic scale. Cancellation between $c = 1$ and the aggregate of the curves $c \geq 2$ is indispensable.

This also shows why a bound of the form

$$\sum_{q \leq Q} e\left(\alpha \frac{q^2 - c^2}{2}\right) \ll Q^{1/2+\varepsilon}$$

cannot hold uniformly in α : at $\alpha = 0$ the left side is $\pi(Q) + O(1)$. The zero and major-arc modes must be cancelled across cofactor curves by the Möbius weights $\mu(c)$ and the exact smooth-transport architecture. Oscillatory estimates are relevant only after that coherent component has been isolated.

Definition 12.2 (Dyadic squared-cofactor exponential sum). For dyadic $C, Q \geq 1$, define

$$F_{C,Q}(\alpha) := \sum_{\substack{C < c \leq 2C \\ \mu(c) \neq 0}} -\mu(c) e\left(-\alpha \frac{c^2}{2}\right) \sum_{\substack{Q < q \leq 2Q \\ q \text{ prime} \\ q > P^+(c)}} e\left(\alpha \frac{q^2}{2}\right). \quad (133)$$

A concrete analytic route is a major/minor-arc decomposition for $F_{C,Q}$, coupled to the Fourier representation of localized squared-space kernels. The major arcs must reproduce and cancel the coherent cofactor-curve main terms, while the minor arcs must satisfy a large-sieve or mean-square estimate. This is the analytic content that the conformal geometry identifies but does not itself prove.

13 Vertical spacing on a fixed real fibre

Fix $m > 0$ and parameterize the full factor hyperbola by signed $b \in \mathbb{R}$:

$$a = \sqrt{m + b^2}, \quad Y_m(b) = 2b\sqrt{m + b^2}. \quad (134)$$

Then

$$Y'_m(b) = 2 \frac{m + 2b^2}{\sqrt{m + b^2}} \geq 2\sqrt{m}. \quad (135)$$

The derivative is positive for all signed b , so the map is strictly increasing from the lower to the upper half-plane.

If admissible b values differ by at least h , then

$$|\Delta Y| \geq 2h\sqrt{m}. \quad (136)$$

At square scale $m \asymp n^2$, vertical spacing is at least order n . This agrees with the conformal scale factor in [Equation \(129\)](#).

14 The square-prefix variance as a squared-space kernel

For a canonical point w_m , define

$$p_n(w_m) := \mathbf{1}_{\{e(m) \leq n\}}. \quad (137)$$

Then

$$S_n = \int p_n(w) d\nu(w). \quad (138)$$

For a terminal index N , define

$$K_N(w, w') := \sum_{n=1}^N p_n(w) p_n(w'). \quad (139)$$

Since each p_n is a tail indicator,

$$K_N(w, w') = (N - \max\{e(w), e(w')\} + 1)_+. \quad (140)$$

Therefore

$$\boxed{\sum_{n=1}^N |S_n|^2 = \iint_{\mathcal{W} \times \mathcal{W}} K_N(w, w') d\nu(w) d\nu(w').} \quad (141)$$

The channel indicators satisfy

$$p_n = a_n + u_n, \quad (142)$$

so the kernel splits exactly into smooth–smooth, transport–transport, and two cross terms.

For transport, the overlap kernel has the interval form

$$K_N^{UU}(w, w') = \#([e(w), h(w)) \cap [e(w'), h(w')] \cap [1, N]). \quad (143)$$

Consequently,

$$K_N^{UU}(w, w') \leq \min\{L(w), L(w')\}. \quad (144)$$

The imaginary coordinate therefore controls temporal support through [Equation \(107\)](#).

15 A rigorous cubic bound for the low-height sector

Fix $\Lambda > 0$. Assign each source m to its entry block

$$j = e(m) = \lfloor \sqrt{m} \rfloor \quad (145)$$

and define the low-height block increment

$$d_j^{\text{low}} := \sum_{\substack{m \in B_j \\ |Y_m| \leq \Lambda j}} \mu(m). \quad (146)$$

By Equation (117),

$$|d_j^{\text{low}}| \leq \lfloor \Lambda \rfloor. \quad (147)$$

Define

$$S_n^{\text{low}} := \sum_{j \leq n} d_j^{\text{low}}. \quad (148)$$

Even under complete sign coherence,

$$|S_n^{\text{low}}| \leq \lfloor \Lambda \rfloor n. \quad (149)$$

Hence

$$\boxed{\sum_{n \leq N} |S_n^{\text{low}}|^2 \ll_{\Lambda} N^3.} \quad (150)$$

No Möbius cancellation is used.

Let

$$S_n = S_n^{\text{low}} + S_n^{\text{high}}. \quad (151)$$

Then

$$\sum_{n \leq N} |S_n|^2 \leq 2 \sum_{n \leq N} |S_n^{\text{low}}|^2 + 2 \sum_{n \leq N} |S_n^{\text{high}}|^2. \quad (152)$$

Thus the low-height sector is already on the cubic scale. The entire unresolved variance problem may be localized to the high-height point cloud.

16 The high-height cofactor-curve problem

The high sector consists of canonical points satisfying

$$|Y_m| > \Lambda e(m). \quad (153)$$

These points lie far from the real axis relative to their square scale, but they remain highly structured:

1. they lie on the explicit curves γ_c ;
2. their horizontal coordinates are prime samples $x = cq$ with $q > P^+(c)$;
3. their one-dimensional curve spacing is expanded by $\sqrt{c^2 + q^2}$;
4. two-dimensional factor regions are expanded by $c^2 + q^2$;
5. positive-height points have exact transport intervals $[e(m), h(m))$;
6. each curve crosses every moving cutoff exactly once and orthogonally.

For each squarefree cofactor c , define the signed curve measure

$$\nu_c := -\mu(c) \sum_{\substack{q \text{ prime} \\ q > P^+(c)}} \delta_{cq+i(q^2-c^2)/2}. \quad (154)$$

Then

$$\nu = \delta_1 + \sum_{c \geq 1} \nu_c, \quad (155)$$

and the variance decomposes as

$$\sum_{c, c'} \iint K_N(w, w') d\nu_c(w) d\nu_{c'}(w'). \quad (156)$$

The curve $c = 1$ is not a negligible boundary case. By [Equations \(131\) and \(132\)](#), its unsigned energy is of order $N^5 / \log^2 N$. The cubic residual therefore requires near-complete cancellation between this prime curve and the remaining cofactor curves. The numerical results in [Section 19](#) measure that cancellation directly.

16.1 A precise kernel-weighted large-sieve target

Fix a local prefix interval

$$I = [N, N + H), \quad 1 \leq H \leq N, \quad (157)$$

and dyadic ranges $C < c \leq 2C$, $Q < q \leq 2Q$. Let $\omega_{I, \ell}(c, q)$ be a smooth localization to the prefix interval, the transport interval, and a high vertical shell ℓ . Define

$$F_{C, Q, \ell}^I(\alpha) := \sum_{\substack{C < c \leq 2C \\ \mu(c) \neq 0}} -\mu(c) e\left(-\alpha \frac{c^2}{2}\right) \sum_{\substack{Q < q \leq 2Q \\ q \text{ prime} \\ q > P^+(c)}} \omega_{I, \ell}(c, q) e\left(\alpha \frac{q^2}{2}\right). \quad (158)$$

Let $\widehat{K}_{I, \ell}(\alpha) \geq 0$ be the Fourier multiplier of the corresponding localized positive semidefinite prefix/transport-overlap kernel.

For parameters $R_0, \Delta \geq 1$, define the major arcs

$$\mathfrak{M}(R_0, \Delta) := \bigcup_{1 \leq r \leq R_0} \bigcup_{\substack{0 \leq a < r \\ (a, r) = 1}} \left\{ \alpha \in \mathbb{R}/\mathbb{Z} : \left| \alpha - \frac{a}{r} \right| \leq \frac{\Delta}{rN^2} \right\}. \quad (159)$$

For $\alpha = a/r + \beta$ on such an arc, define the explicit Hardy–Littlewood model by replacing the prime indicator in [Equation \(158\)](#) with its reduced-residue main term:

$$\mathcal{M}_{C, Q, \ell}^I(\alpha) := \sum_{\substack{C < c \leq 2C \\ \mu(c) \neq 0}} -\mu(c) e\left(-\alpha \frac{c^2}{2}\right) \frac{G_r(a)}{\varphi(r)} \int_Q^{2Q} \omega_{I, \ell}(c, t) e\left(\beta \frac{t^2}{2}\right) \frac{dt}{\log t}, \quad (160)$$

$$G_r(a) := \sum_{\substack{u \bmod r \\ (u, r) = 1}} e\left(\frac{au^2}{2r}\right). \quad (161)$$

Outside $\mathfrak{M}(R_0, \Delta)$, put $\mathcal{M}_{C, Q, \ell}^I = 0$.

Conjecture 16.1 (Kernel-weighted Squared-Cofactor Large Sieve). *For every $\varepsilon > 0$, there are choices $R_0 = N^{o(1)}$ and $\Delta = N^{o(1)}$ such that, uniformly for $1 \leq H \leq N$,*

$$\sum_{C,Q,\ell} \int_{\mathbb{R}/\mathbb{Z}} |F_{C,Q,\ell}^I(\alpha) - \mathcal{M}_{C,Q,\ell}^I(\alpha)|^2 \widehat{K}_{I,\ell}(\alpha) d\alpha \ll_{\varepsilon} H N^{2+\varepsilon}, \quad (162)$$

$$\int_{\mathbb{R}/\mathbb{Z}} \left| \sum_{C,Q,\ell} \mathcal{M}_{C,Q,\ell}^I(\alpha) \right|^2 \widehat{K}_I(\alpha) d\alpha \ll_{\varepsilon} H N^{2+\varepsilon}. \quad (163)$$

The dyadic sums range only over cells meeting the high sector and the prefix interval, and the localized multipliers satisfy $\sum_{\ell} \widehat{K}_{I,\ell} \ll \widehat{K}_I$.

The first estimate is the kernel-weighted minor-arc and major-arc error bound. The second is the required cancellation of the explicit major-arc models across cofactor scales. Both clauses are necessary: at zero frequency the prime curve is of order $\pi(Q)$, so a uniform pointwise square-root estimate is false. The conjecture places the large-sieve assertion exactly in the norm generated by the square-prefix kernel rather than in an unrelated collision statistic.

Conjecture 16.2 (High cofactor-curve kernel estimate). *After removing the low-height sector controlled by Equation (150), the signed cofactor-curve interactions satisfy, for every $\varepsilon > 0$,*

$$\sum_{n \leq N} |S_n^{\text{high}}|^2 \ll_{\varepsilon, \Lambda} N^{3+\varepsilon}. \quad (164)$$

The two clauses of Conjecture 16.1 are a sufficient analytic route: the first controls the localized remainder and the second controls the cofactor-cancelled major-arc model.

This formulation is stronger than rewriting RH in the coordinate Y . It asks for separate control of cofactor ranges, prime ranges, squared-space shells, and transport-overlap multipliers. It also identifies the precise analytic obstruction: a one-dimensional prime-sampled large sieve coupled across the Möbius-weighted family of cofactor parabolas.

17 Exponent accounting in squared complex space

The block increment

$$d_n = S_n - S_{n-1} \quad (165)$$

satisfies the trivial bound $|d_n| \leq 2n + 1$. Expanding the cumulative energy gives

$$\sum_{n \leq N} |S_n|^2 = \sum_{j,k \leq N} (N - \max\{j, k\} + 1) d_j d_k. \quad (166)$$

The fully coherent ceiling is

$$N^5 = \underbrace{N}_{\text{prefix multiplicity}} \times \underbrace{N^2}_{\text{block pairs}} \times \underbrace{N^2}_{\text{raw pair amplitude}}. \quad (167)$$

The squared map provides the exact missing scale:

$$J(c, q) = c^2 + q^2 \geq 2cq \asymp n^2. \quad (168)$$

Thus a unit area of factor combinations is diluted by n^{-2} in the squared plane. Equivalently, each coordinate direction is stretched by order n . If the localized kernel respects this conformal dilation, then the effective pair amplitude becomes order one:

$$N^3 = \underbrace{N}_{\text{prefix multiplicity}} \times \underbrace{N^2}_{\text{block pairs}} \times \underbrace{1}_{\text{squared-space interaction}}. \quad (169)$$

The low-height theorem proves this reduction directly in the region $|Y| \lesssim n$. The high-height conjecture asks for the corresponding operator statement in the remaining region.

18 A uniform local criterion for RH

The global cubic energy is the natural benchmark, but a fully pointwise converse is obtained from a uniform local estimate. Define

$$V_{\text{loc}}(N, H) := \sum_{n=N}^{N+H-1} |S_n|^2. \quad (170)$$

Theorem 18.1 (Uniform local square-prefix criterion). *The Riemann Hypothesis is equivalent to the statement that, for every $\varepsilon > 0$,*

$$V_{\text{loc}}(N, H) \ll_{\varepsilon} H N^{2+\varepsilon} \quad (1 \leq H \leq N). \quad (171)$$

Proof. Under RH, the classical Mertens criterion gives

$$M(x) \ll_{\delta} x^{1/2+\delta}$$

for every $\delta > 0$. Since $X_n \asymp n^2$, one has

$$|S_n|^2 \ll_{\varepsilon} n^{2+\varepsilon},$$

and summing H consecutive terms gives [Equation \(171\)](#).

Conversely, set $H = 1$. Then

$$|S_N|^2 \ll_{\varepsilon} N^{2+\varepsilon},$$

so $|S_N| \ll_{\eta} N^{1+\eta}$ for every $\eta > 0$. If

$$n^2 \leq x < (n+1)^2,$$

then

$$|M(x) - M(n^2 - 1)| \leq 2n + 1$$

because $|\mu(m)| \leq 1$. Therefore

$$M(x) \ll_{\eta} x^{1/2+\eta},$$

which is equivalent to RH. □

A local version of [Theorem 16.2](#) would therefore complete the argument without relying on a global-average converse.

19 Numerical results and pre-registered diagnostics

The computations reported here use exact integer arithmetic through

$$N_{\max} = 4096, \quad X_{\max} = (4097)^2 - 1 = 16,785,408. \quad (172)$$

A linear sieve produced

$$1,078,362 \text{ primes} \quad \text{and} \quad 10,204,280 \text{ squarefree sources } m \geq 2. \quad (173)$$

19.1 Exact checks

The implementation independently constructed S_n , A_n , and T_n from the source entry and transition intervals. Over all 4096 prefixes,

$$\max_{n \leq 4096} |S_n - (A_n - T_n)| = 0. \quad (174)$$

Across all 10,204,280 squarefree sources, the following violation counts were also zero:

Table 2: Exact squared-space verification through $X_{\max}$.

Check	Violations
$(q^2 + c^2)^2 = (2m)^2 + (q^2 - c^2)^2$	0
$c^2(q^2 - c^2) = m^2 - c^4$	0
height–lifetime inequality	0
prefix identity $S_n = A_n - T_n$	0

19.2 Smooth–transport matching

Table 3: Exact cumulative prefix energies. Every entry is computed from all prefixes $1 \leq n \leq N$.

N	$\ A\ ^2/N^3$	$\ T\ ^2/N^3$	$\langle A, T \rangle/N^3$	$\ A - T\ ^2/N^3$	S_N
128	0.490490	0.469682	0.475665	0.008842	−18
256	1.386025	1.338898	1.356626	0.011671	4
512	3.709722	3.668905	3.685593	0.007442	38
1024	9.831997	9.807089	9.814616	0.009853	288
2048	26.661079	26.658451	26.655516	0.008497	263
4096	72.821788	73.064243	72.938028	0.009974	329

The residual remains near the cubic scale while the two component energies increase substantially.

19.3 The prime curve does not satisfy the cubic bound

Let

$$S_n^{(1)} = -\pi(X_n)$$

be the contribution of the curve $c = 1$, and put

$$S_n^{(\geq 2)} = S_n - S_n^{(1)}.$$

Table 4: Prime-curve versus remaining-curve energy decomposition.

N	V/N^3	$V_{c=1}/N^3$	$V_{c \geq 2}/N^3$	$2\langle c = 1, c \geq 2 \rangle/N^3$	$V/V_{c=1}$	correlation
128	0.008842	51.225450	50.965559	-102.182168	1.7260×10^{-4}	-0.999916714
256	0.011671	146.590116	146.035583	-292.614028	7.9618×10^{-5}	-0.999961911
512	0.007442	441.694929	441.211486	-882.898974	1.6848×10^{-5}	-0.999991721
1024	0.009853	1382.291036	1381.948964	-2764.230146	7.1282×10^{-6}	-0.999996443
2048	0.008497	4450.731250	4450.668331	-8901.391084	1.9091×10^{-6}	-0.999999045
4096	0.009974	14652.537064	14655.994904	-29308.521994	6.8072×10^{-7}	-0.999999667

Then

$$\|S\|^2 = \|S^{(1)}\|^2 + \|S^{(\geq 2)}\|^2 + 2\langle S^{(1)}, S^{(\geq 2)} \rangle. \quad (175)$$

At $N = 4096$, the prime curve has energy more than $1.46 \times 10^4 N^3$, while the total residual is approximately $9.97 \times 10^{-3} N^3$. The squared norm of the total is only 6.81×10^{-7} of the prime-curve squared norm. This is direct numerical evidence that the central phenomenon is cancellation *between* cofactor curves, not cubic behavior of the prime curve in isolation.

19.4 The dyadic cofactor cascade

For $C \geq 1$, define the partial cofactor residual

$$R_C(x) := 1 - \sum_{1 \leq c \leq C} \mu(c) \Pi_c(x), \quad (176)$$

and its square-prefix energy

$$V_C(N) := \sum_{n \leq N} |R_C(X_n)|^2. \quad (177)$$

Thus

$$R_1(x) = 1 - \pi(x),$$

and $R_C(x) = M(x)$ once C contains every canonical cofactor that can occur below x . The cascade tests how successive dyadic cofactor ranges cancel the prime curve.

The cascade is strongly nonmonotone. The first four cofactor scales reduce the normalized energy from 14652.49 at $C = 1$ to 0.1485 at $C = 16$, but the next dyadic block raises it to 344.81 at $C = 32$. Subsequent blocks repeatedly destroy and recreate coherent mass before the residual settles near the final cubic scale. This rules out a simple monotone “more cofactors means more cancellation” picture. The analytic estimate must control a signed cascade of dyadic cross-terms.

The machine-readable file `cofactor_cascade.csv` reports every dyadic C for all six prefix checkpoints $N = 128, 256, 512, 1024, 2048, 4096$.

19.5 The dyadic Gram matrix and the $16 < c \leq 32$ spike

Let B_j denote the complete square-prefix vector contributed by the cofactor block

$$2^{j-1} < c \leq 2^j,$$

with $B_0 = P$ for $c = 1$. The exact Gram matrix is

$$G_{ij}(N) := \langle B_i, B_j \rangle_{1 \leq n \leq N}. \quad (178)$$

Table 5: Complete dyadic cofactor cascade at $N = 4096$. The third column is $V_C(4096)/4096^3$; the fourth is V_C/V_1 . The final row is exact because the largest observed canonical cofactor is $692,835 < 2^{20}$.

C	$R_C(X_{4096})$	V_C/N^3	V_C/V_1
1	-1,078,361	14652.492160	1.000e+00
2	-513,936	3318.367550	2.265e-01
4	-127,056	198.352150	1.354e-02
8	86,580	99.704527	6.805e-03
16	-4,620	0.148513	1.014e-05
32	162,415	344.808298	2.353e-02
64	51,754	36.470907	2.489e-03
128	75,937	76.897452	5.248e-03
256	67,189	59.850472	4.085e-03
512	81,166	84.906888	5.795e-03
1,024	71,179	60.275193	4.114e-03
2,048	43,260	15.704342	1.072e-03
4,096	3,633	4.361902	2.977e-04
8,192	-39,326	14.649760	9.998e-04
16,384	-15,331	1.552681	1.060e-04
32,768	-2,326	0.051346	3.504e-06
65,536	4,622	0.142635	9.735e-06
131,072	708	0.011615	7.927e-07
262,144	206	0.009590	6.545e-07
524,288	332	0.009980	6.811e-07
1,048,576	329	0.009974	6.807e-07

The following table compares the harmonic Möbius coefficient

$$a_j = \sum_{2^{j-1} < c \leq 2^j} \frac{\mu(c)}{c}$$

from [Theorem 2.5](#) with the finite-sample regression coefficient

$$\hat{\beta}_j := \frac{\langle P, B_j \rangle}{\|P\|^2}.$$

Table 6: Dyadic cofactor block geometry at $N = 4096$. The close agreement of signs, the nearly unit correlations, and the Gram spectrum below exhibit the major-arc rank-one structure.

block	a_j	$\hat{\beta}_j$	$\text{corr}(P, B_j)$	$\ B_j\ ^2/N^3$	$\Delta V/N^3$
$1 < c \leq 2$	-0.500000	-0.524110	-0.999999	4024.9308	-11334.1246
$2 < c \leq 4$	-0.333333	-0.359551	-0.999997	1894.2410	-3120.0154
$4 < c \leq 8$	-0.176190	-0.198795	-0.999993	579.0658	-98.6476
$8 < c \leq 16$	0.070263	0.085079	0.999979	106.0651	-99.5560
$16 < c \leq 32$	-0.123472	-0.155995	-0.999970	356.5833	344.6598
$32 < c \leq 64$	0.078716	0.103523	0.999952	157.0458	-308.3374

For the block $B_{16,32}$ one finds

$$\frac{\|B_{16,32}\|^2}{N^3} = 356.583335, \quad (179)$$

$$\frac{\langle P, B_{16,32} \rangle}{N^3} = -2285.725157, \quad (180)$$

$$\text{corr}(P, B_{16,32}) = -0.999970. \quad (181)$$

Thus the proposed balance $\langle P, B_{16,32} \rangle \approx -\frac{1}{2}\|B_{16,32}\|^2$ is not what occurs. Instead,

$$\boxed{\langle P, B_{16,32} \rangle = -6.410073 \|B_{16,32}\|^2 \quad \text{at } N = 4096.} \quad (182)$$

The block is almost a negative scalar multiple of the prime curve, as predicted by the rank-one theorem.

The cascade spike has a different cause. Let R_{16} be the cumulative residual after the blocks through $c \leq 16$. Then

$$\frac{\langle R_{16}, B_{16,32} \rangle}{N^3} = -5.961775, \quad (183)$$

so

$$\frac{V_{32} - V_{16}}{N^3} = \frac{\|B_{16,32}\|^2 + 2\langle R_{16}, B_{16,32} \rangle}{N^3} \quad (184)$$

$$= 356.583335 - 11.923550 \quad (185)$$

$$= 344.659785. \quad (186)$$

By $C = 16$, the prime-like coherent mode has already been almost annihilated. The block $16 < c \leq 32$ therefore reintroduces that mode rather than cancelling it. The next block $32 < c \leq 64$ has

$$\frac{\langle R_{32}, B_{32,64} \rangle}{N^3} = -232.691603$$

and lowers the energy by $308.337391N^3$, repairing most of the spike.

The Gram matrix itself is nearly rank one. For the six blocks through $c \leq 32$, its largest eigenvalue contains

$$0.999998101 \quad (187)$$

of the trace and $\lambda_2/\lambda_1 = 1.8926 \times 10^{-6}$. Through $c \leq 64$, the corresponding values are 0.999997519 and 2.4696×10^{-6} . Restricting to the high sector $|Y| > 16n$ changes these figures only to 0.999998118 and 1.8753×10^{-6} through $c \leq 32$. The observed rank-one mode is therefore a high-sector phenomenon, not an artifact of the already controlled low-height points.

19.6 Low-height clustering and lifetime

For fixed Λ , define

$$C_n(\Lambda) := \#\{w_m \in \mathcal{B}_n : |Y_m| \leq \Lambda n\}, \quad (188)$$

$$I_n(\Lambda) := \sum_{\substack{w_m \in \mathcal{B}_n \\ |Y_m| \leq \Lambda n}} (1 + L(m)). \quad (189)$$

The sharpened exact bounds are

$$C_n(\Lambda) \leq \lfloor \Lambda \rfloor, \quad (190)$$

$$I_n(\Lambda) \leq \lfloor \Lambda \rfloor (1 + 2\Lambda), \quad (191)$$

with $C_n(1) = I_n(1) = 0$. The odd and even source counts are bounded by $\lfloor \Lambda/2 \rfloor$ and $\lceil \lfloor \Lambda \rfloor / 2 \rceil$, respectively.

The zero at $\Lambda = 1$ is now explained exactly by [Equation \(113\)](#); it is not merely an empirical absence. The observed total and parity occupancies remain substantially below the sharpened deterministic envelopes.

Table 7: Observed low-height maxima over $128 \leq n \leq 4096$, with the sharpened general and parity bounds.

Λ	count bound	total max	odd bound	odd max	even bound	even max	incidence bound	incidence max
1	0	0	0	0	0	0	0	0
2	2	1	1	0	1	1	10	1
4	4	2	2	1	2	1	36	4
8	8	4	4	2	4	2	136	10
16	16	6	8	3	8	3	528	26

19.7 Uniform local ratios

For a dyadic base N , define

$$\mathfrak{R}(N) := \max_{\substack{N \leq M \leq 2N-H \\ H \in \{1,2,4,\dots,N\}}} \frac{\sum_{n=M}^{M+H-1} |S_n|^2}{HM^2}. \quad (192)$$

Table 8: Worst local ratios from the exact range $n \leq 4096$.

base N	$\mathfrak{R}(N)$	$\mathfrak{R}(N)/N^{0.10}$	maximizing start	maximizing H
128	0.176193	0.108459	243	1
256	0.145830	0.083757	309	1
512	0.183003	0.098069	547	1
1024	0.157837	0.078918	1032	1
2048	0.166479	0.077665	2544	1

The fitted log–log slope of $\mathfrak{R}(N)$ over these five dyadic scales is

$$\hat{\theta} = -0.00494746. \quad (193)$$

This range passes the preliminary scale-stability test $\hat{\theta} \leq 0.10$, but it is not long enough to determine the asymptotic law.

19.8 What remains to be computed

The next numerical stage is not another aggregate Mertens table. It must resolve the high-sector kernel by dyadic cofactor ranges, prime ranges, height shells, and Fourier frequency. For every (C, Q, ℓ) cell it should report:

1. the curve-sample count and one-dimensional arclength density;
2. the zero-mode curve mass;
3. the major-arc approximation and its error;
4. the minor-arc mean square of $F_{C,Q}$;
5. the corresponding localized kernel row and column sums;
6. the contribution of the cell to $\langle \nu_C, K_N \nu_{C'} \rangle$;

7. the cancellation of $c = 1$ against successive cofactor ranges.

The pre-registered high-sector pass condition is that, after explicit major-arc subtraction, the normalized minor-arc mean squares and localized kernel row sums remain bounded by N^ε across successive dyadic scales. Failure of that condition falsifies the proposed Squared-Cofactor Large Sieve route even if the aggregate residual continues to look cubic.

20 Lean formalization architecture

The exact development separates the arithmetic decomposition from the open high-sector estimate.

20.1 Factor and squared coordinates

1. `Geometry/FullFactorFiber.lean`: all positive factor pairs of N , Fermat coordinates, and the full and nontrivial envelopes.
2. `Geometry/CanonicalFactor.lean`: $q = P^+(m)$, $c = m/q$, signed b , and signed Y .
3. `Geometry/ComplexSquare.lean`: $(a + bi)^2 = m + iY$.
4. `Geometry/SquaredRecovery.lean`: $c^2 = |w| - Y$ and $q^2 = |w| + Y$.
5. `Geometry/NullCoordinates.lean`: $U = c^2$, $V = q^2$, and $x^2 = UV$.

20.2 Cofactor curves and cutoffs

1. `Geometry/CofactorParabola.lean`: define γ_c and prove that canonical points lie on it.
2. `Geometry/MovingCutoff.lean`: define Γ_R and the squared-space smooth and transport regions.
3. `Geometry/CurveIntersection.lean`: prove [Equations \(71\)](#), [\(72\)](#) and [\(77\)](#).
4. `Geometry/ConformalJacobian.lean`: prove [Equations \(125\)](#), [\(127\)](#) and [\(128\)](#).

20.3 Telescoping and clustering

1. `Kernel/EntryTransition.lean`: define $e(m)$, $h(m)$, and $L(m)$.
2. `Kernel/MovingRegionBridge.lean`: prove $\mathcal{A}(R_n) = A_n$ and $\mathcal{U}(R_n) = -T_n$.
3. `Kernel/ChannelTelescoping.lean`: prove [Equations \(94\)](#) to [\(96\)](#).
4. `Geometry/HeightLifetime.lean`: prove the exact identity [Equation \(102\)](#) and the discrete bound [Equation \(106\)](#).
5. `Geometry/AbsoluteClustering.lean`: prove [Theorems 11.1](#) and [11.2](#) and [Equation \(120\)](#).
6. `Kernel/LowHeightCubic.lean`: formalize [Equation \(150\)](#).

20.4 Open input

The unresolved statement should be a named proposition and passed as a hypothesis:

```
def HighCofactorCurveKernelStatement : Prop :=
  forall eps > 0, exists C,
    forall N,
      highEnergy N <= C * N ^ (3 + eps)

theorem rh_of_highCofactorCurveKernel
  (h : HighCofactorCurveKernelStatement) :
  RHStatement := by
  ...
```

No exact geometry module should import this open statement.

21 Logical status and remaining proof obligation

Table 9: Status of the squared complex factor-space program.

Statement	Status
Exact square-root smooth-transport decomposition	Proved
Full factor-fibre intervals and envelopes	Proved
Canonical largest-prime point selection	Proved
Signed squared coordinate and factor recovery	Proved
Null coordinates $U = c^2$, $V = q^2$	Proved
Cofactor parabola formula	Proved
Moving cutoff parabola	Proved
Exact and orthogonal curve intersection	Proved
Moving-region form of $A - T$	Proved
Channel telescoping	Proved
Exact signed height-lifetime identity	Proved
Absolute low-height clustering	Proved
Low-height cubic energy bound	Proved
Conformal Jacobian and two-power expansion	Proved
High cofactor-curve signed kernel estimate	Open
Unconditional proof of RH	Not claimed

The open theorem is no longer an unspecified cancellation assertion. It is the demand that the exact prefix kernel, restricted to the high squared-space sector and decomposed by cofactor curves, inherit the n^{-2} area dilution of the conformal map together with the exact temporal-support bounds.

22 Conclusion

The squared complex plane unifies the factorization geometry and the square-prefix smooth-transport architecture.

The full factor fibre of N is the vertical line

$$\operatorname{Re} w = N,$$

with each factor pair encoded by

$$w = N + i \frac{v^2 - u^2}{2}.$$

The original factor-search bound with smallest odd factor 3 describes the nontrivial envelope, while the complete fibre begins at factor 1 and reaches

$$Y_{\max} = \frac{N^2 - 1}{2}.$$

The Möbius problem selects one canonical point per squarefree integer:

$$q = P^+(m), \quad c = m/q, \quad w_m = m + i \frac{q^2 - c^2}{2}.$$

These points lie on the discrete family of cofactor parabolas

$$\gamma_c(x) = \frac{x^2 - c^4}{2c^2}.$$

At cutoff R , the complete prefix is the vertical strip $x < R^2$, and the smooth-transport split is the moving parabola

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}.$$

Each γ_c intersects Γ_R exactly and orthogonally at $x = cR$, which is precisely the transition $q = R$.

The point enters the prefix at $R = \sqrt{cq}$ and crosses into smoothness at $R = q$. The crossing mass telescopes between $-T$ and A . The imaginary coordinate is the exact transition clock:

$$\tau(c, q) = \frac{2Y}{q(1 + \sqrt{c/q})(1 + c/q)}.$$

Low absolute height forces small factor gap, bounded block occupancy, and bounded transport lifetime. This sector already satisfies a cubic energy bound without using sign cancellation.

The quadratic map itself supplies the missing scale:

$$\det D\Phi(c, q) = c^2 + q^2 \geq 2cq.$$

At square scale $cq \asymp n^2$, the factor lattice is expanded by order n in each direction and by order n^2 in area. This is exactly the two-power geometric dilution required by the variance accounting.

What remains is a sharply localized theorem: prove that the high, signed, prime-sampled cofactor-curve interactions inherit that conformal dilution inside the exact square-prefix kernel. Establishing that operator estimate would complete the cubic residual bound and, in its uniform local form, imply the Riemann Hypothesis.

Symbol	Meaning
B_n	square block $[n^2, (n+1)^2)$
R_n	square-root cutoff $n+1$
X_n	complete-prefix endpoint $R_n^2 - 1$
S_n	$M(X_n)$
A_n	signed R_n -smooth mass
T_n	one-large-prime cofactor transport statistic
c	canonical cofactor $m/P^+(m)$
q	canonical largest prime $P^+(m)$
a, b	signed Fermat coordinates $(c+q)/2, (q-c)/2$
$w = x + iy$	squared point $cq + i(q^2 - c^2)/2$
$Y = y$	signed ordinate $2ab$
ρ	$ w = (c^2 + q^2)/2$
U, V	null coordinates c^2, q^2
γ_c	fixed cofactor parabola
Γ_R	moving smooth-transport cutoff parabola
$e(m)$	first complete-prefix index containing m
$h(m)$	smoothness-transition index $q-1$
$L(m)$	discrete transport lifetime
$\tau(c, q)$	signed continuous delay $q - \sqrt{cq}$
K_N	cumulative square-prefix kernel

A Notation summary

B Reproducibility protocol

The numerical tables in [Section 19](#) are reproduced by the standalone script

```
python squared_space_reproducibility_v3.py \  
  --nmax 4096 \  
  --out squared_space_repro_v2
```

using Python, NumPy, and Numba. No network data, probabilistic sampling, or floating-point primality test is used.

B.1 Deterministic algorithm

1. Set

$$X_{\max} = (N_{\max} + 1)^2 - 1.$$

2. Run a linear sieve through $X_{\max}$ to compute:

$$\mu(m), \quad \text{spf}(m), \quad P^+(m), \quad \pi(m).$$

3. Form

$$S_n = M((n + 1)^2 - 1)$$

by integer prefix summation.

4. Construct A_n and T_n independently from the source entry/transition intervals:

$$e(m) = \lfloor \sqrt{m} \rfloor, \quad h(m) = P^+(m) - 1.$$

A squarefree source contributes to T on $[e, h)$ and to A on $[\max\{e, h\}, N_{\max}]$.

5. For every squarefree $m > 1$, put

$$q = P^+(m), \quad c = m/q, \quad 2Y = q^2 - c^2.$$

Verify the norm, curve, and lifetime identities using integers only.

6. Compute the cofactor-1 curve exactly as

$$S_n^{(1)} = -\pi(X_n),$$

then set

$$S_n^{(\geq 2)} = S_n - S_n^{(1)}$$

and evaluate all three terms in [Equation \(175\)](#).

7. Bin every squarefree source by the least dyadic threshold $C = 2^j$ containing its canonical cofactor $c = m/P^+(m)$. Cumulatively sum the bins to obtain $R_C(X_n)$ and $V_C(N)$ for every dyadic C and every reported N .
8. Evaluate the total, odd, and even low-height counts and incidences for $\Lambda = 1, 2, 4, 8, 16$.
9. Evaluate $\mathfrak{R}(N)$ for dyadic bases 128, 256, 512, 1024, 2048 and all dyadic window lengths $H \leq N$.

B.2 Generated output files

The script writes:

Table 10: Machine-readable reproducibility outputs.

File	Contents
<code>run_summary.txt</code>	range, source counts, exact violation counts, and local-ratio slope
<code>prefix_energy.csv</code>	S_N , total energy, and A - T component energies
<code>prime_curve_decomposition.csv</code>	$c = 1$, $c \geq 2$, cross term, correlation, and residual ratio
<code>cofactor_cascade.csv</code>	$R_C(X_N)$ and $V_C(N)$ for every dyadic C and reported N
<code>low_height_diagnostics.csv</code>	sharpened total/parity bounds and observed clustering/lifetime maxima
<code>local_ratio_diagnostics.csv</code>	worst local ratios and maximizing intervals

For the reported run, the summary must read:

```

N_max: 4096
X_max: 16785408
prime_count: 1078362
squarefree_sources_m_ge_2: 10204280
max_observed_canonical_cofactor: 692835
max_dyadic_cofactor_threshold: 16777216
max_abs_S_minus_A_plus_T: 0
nonzero_prefix_identity_errors: 0
norm_identity_violations: 0
cofactor_curve_identity_violations: 0
height_lifetime_violations: 0
cofactor_cascade_full_error: 0
local_ratio_log_log_slope: -0.004947461592

```

B.3 Integrity hashes

The SHA-256 hashes of the revised script and machine-readable outputs are listed in the release archive and in the accompanying `SHA256SUMS.txt`. This avoids copying stale hashes into the manuscript when the numerical protocol is extended.

The exact checks are binary: any nonzero violation invalidates the run. The asymptotic diagnostics are not proofs; their purpose is to identify which geometric or analytic component fails first as the range is extended.

References

- [1] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.
- [2] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge University Press, 2007.

- [3] T. Tao and V. Vu, *Additive Combinatorics*, Cambridge University Press, 2006.
- [4] F. Viole, *Complex Space Factorization*, working paper.